Apprentice Aptitude Battery

Practice Problems

GAN HR

Copyright: GAN Human Resources 2023



GENERAL DIRECTIONS

This booklet includes practice problems for each of the four GAN Aptitude Tests.

You should study the GAN Study Guide before attempting the practice problems. The study guide provides an explanation of how to take each test.

Each section for these practice problems has a recommended time limit.

NUMERICAL COMPUTATION: 30 minutes

1. 1600.4 ÷ 4 =	(A) 400.3	(B) 200.4	(C) 100.1	(D) 400.1	(E) 300.2
2. 41.9 – 3.84 =	(A) 38.00	(B) 28.06	(C) 28.00	(D) 38.2	(E) 38.06
3. $^{1}/_{8}$ x $^{3}/_{5}$ =	(A) $^{2}/_{40}$	(B) $^{3}/_{40}$	(C) $^{5}/_{24}$	(D) $^{1}/_{40}$	(E) $^{2}/_{80}$
4. 78 - 64 =	(A) 13	(B) 28	(C) 8	(D) 14	(E) 6
5. $^{1}/_{4} + ^{2}/_{3} + ^{1}/_{6}$ =	(A) $2^{1}/_{12}$	(B) $^{1}/_{18}$	(C) $^{1}/_{12}$	(D) $1^{1}/_{24}$	(E) $1^{1}/_{12}$
6. 11.322 – 7.888 =	(A) 3.343	(B) 1.333	(C) 3.434	(D) 3.833	(E) 2.433
7. 184.32 ÷ 8 =	(A) 23.04	(B) 66.08	(C) 33.08	(D) 30.04	(E) 53.2
8. $^{2}/_{3}$ - $^{1}/_{8}$ =	(A) $^{1}/_{24}$	(B) $^{26}/_{24}$	(C) $^{8}/_{24}$	(D) $^{1}/_{24}$	(E) $^{13}/_{24}$
9. 52.72 ÷ 4 =	(A) 13.18	(B) 26.18	(C) 2.18	(D) 18.13	(E) 18.26
10. 19.4 – 6.32 =	(A) 13.08	(B) 13.18	(C) 26.16	(D) 13.14	(E) 13.10
11. 31.6 ÷ 4 =	(A) .099	(B) .99	(C) 7.9	(D) 18.9	(E) 21.2
12. $7\ ^{1}/_{5} + 4\ ^{3}/_{4}$ =	(A) $11\ ^{1}/_{19}$	(B) $11\ ^{19}/_{20}$	(C) $10\ ^{19}/_{20}$	(D) $13\ ^{1}/_{20}$	(E) $10\ ^{3}/_{19}$
13. 34% of 20 =	(A) 6.80	(B) 16.6	(C) 68.0	(D) .680	(E) .166
14. $7\ ^{1}/_{9} + 5\ ^{1}/_{6}$ =	(A) $12\ ^{3}/_{4}$	(B) $12\ ^{1}/_{9}$	(C) $13\ ^{5}/_{18}$	(D) $12\ ^{3}/1_{8}$	(E) $12\ ^{5}/_{18}$
15. $^{1}/_{5} \div ^{7}/_{10}$ =	(A) $^{1}/_{7}$	(B) $^{4}/_{14}$	(C) $2\ ^{3}/_{7}$	(D) $^{2}/_{7}$	(E) $^{1}/_{8}$
16. $^{1}/_{2} + ^{1}/_{3} + ^{1}/_{9}$ =	(A) $1^{1}/_{18}$	(B) $^{17}/_{18}$	(C) $^{16}/_{18}$	(D) $^{34}/_{36}$	(E) $^{1}/_{18}$
17. (25 ÷ $^{1}/_{5}$) – 4 =	(A) 122	(B) 24	(C) 20	(D) 121	(E) 21
18. (125 ÷ 25) + 4.9 =	(A) 11.1	(B) 10.9	(C) 18.9	(D) 9.0	(E) 9.9
19. $5\ ^{1}/_{4} – 3\ ^{2}/_{5}$ =	(A) $1\ ^{17}/_{20}$	(B) $1\ ^{1}/_{20}$	(C) $1\ ^{34}/_{40}$	(D) $2\ ^{17}/_{20}$	(E) $^{17}/_{20}$
20. $3\ ^{1}/_{8}$ x $2\ ^{1}/_{3}$ =	(A) $6^{5}/_{7}$	(B) $5^{7}/_{12}$	(C) $6^{7}/_{24}$	(D) $7^{7}/_{24}$	(E) $7^{1}/_{24}$
21. 55 – 8 – 6 =	(A) 38	(B) 40	(C) 42	(D) 51	(E) 41
22. $^{1}/_{6} \div ^{2}/_{11}$ =	(A) $^{11}/_{12}$	(B) $1\ ^{1}/_{12}$	(C) $^{9}/_{12}$	(D) $2\ ^{11}/_{12}$	(E) $1\ ^{11}/_{12}$

23. $22.54 \div 2.3 =$ (A) 10.8 (B) 9.8 (C) 3.8 (D) 11.8 (E) 8.8

24. $2\ ^{2}/_{3} \div\ ^{1}/_{5} =$ (A) $12\ ^{1}/_{8}$ (B) $13\ ^{3}/_{8}$ (C) $13\ ^{2}/_{3}$ (D) $13\ ^{1}/_{3}$ (E) $14\ ^{5}/_{16}$

25. 18% of 50 = (A) 6 (B) 11 (C) 12 (D) 10 (E) 9

26. $6.22 + (4.4 \times\ ^{1}/_{4}) =$ (A) 7.12 (B) 7.41 (C) 6.12 (D) 7.32 (E) 7.12

27. $3\ ^{1}/_{5} - 2\ ^{5}/_{6} =$ (A) $^{9}/_{30}$ (B) $^{8}/_{11}$ (C) $^{11}/_{30}$ (D) $^{10}/_{20}$ (E) $^{10}/_{30}$

28. 9% of 11 = (A) .98 (B) .99 (C) 1.08 (D) .90 (E) .89

29. $31.6 \times 3.06 =$ (A) 96.696 (B) 86.796 (C) 94.966 (D) 96.496 (E) 94.696

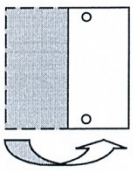
30. 5.4% of 18 = (A) .839 (B) .894 (C) .927 (D) .984 (E) .972

NUMERICAL REASONING: 22 minutes

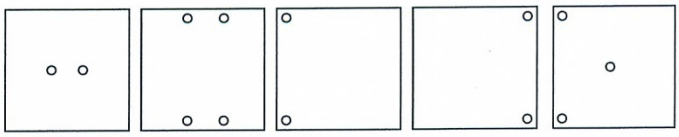
1. 8 10 14 20 28	(A) 37	(B) 38	(C) 36	(D) 35	(E) 39
2. 34 33 31 28 24	(A) 21	(B) 20	(C) 17	(D) 18	(E) 19
3. 96 92 91 87 86	(A) 80	(B) 84	(C) 83	(D) 82	(E) 81
4. 14 16 20 26 34	(A) 44	(B) 43	(C) 42	(D) 41	(E) 40
5. 45 49 57 69 85	(A) 101	(B) 100	(C) 95	(D) 115	(E) 105
6. 8 12 24 28 56	(A) 112	(B) 61	(C) 60	(D) 62	(E) 110
7. 1701 567 189 63 21	(A) 6	(B) 7	(C) 5	(D) 8	(E) 10
8. 75 65 57 51 47	(A) 45	(B) 46	(C) 44	(D) 43	(E) 48
9. 63 55 48 42 37 33	(A) 33	(B) 29	(C) 32	(D) 30	(E) 31
10. 44 22 24 12 14	(A) 8	(B) 7	(C) 9	(D) 6	(E) 28
11. $^1/_{16,807}$ $^1/_{2,401}$ $^1/_{343}$ $^1/_{49}$ $^1/_7$	(A) $^1/_{49}$	(B) 1	(C) 0	(D) $^1/_7$	(E) 10
12. 1 2 4 7 28	(A) 102	(B) 33	(C) 56	(D) 34	(E) 32
13. 88 44 46 23 25	(A) 13 $^1/_2$	(B) 27	(C) 11 $^1/_2$	(D) 21	(E) 12 $^1/_2$
14. 42 37 47 42 52	(A) 46	(B) 48	(C) 57	(D) 47	(E) 42
15. 7 10 30 33 99	(A) 33	(B) 95	(C) 102	(D) 100	(E) 297
16. 28672 3584 448 56 7	(A) $^7/_8$	(B) 1 $^1/_8$	(C) $^1/_2$	(D) $^3/_4$	(E) $^8/_7$
17. $^6/_{125}$ $^6/_{25}$ $^6/_5$ 6 30	(A) 120	(B) 150	(C) 100	(D) 1	(E) 14
18. $^1/_{21}$ $^1/_7$ $^2/_7$ $^{10}/_{21}$ $^5/_7$	(A) $^3/_{14}$	(B) $^{30}/_{35}$	(C) 2	(D) 7	(E) 1
19. 13824 1728 288 48 8	(A) 2 $^1/_3$	(B) 1 $^1/_4$	(C) 1 $^4/_5$	(D) 1 $^1/_3$	(E) 4
20. $^1/_{12}$ $^1/_4$ 1 5 30	(A) 110	(B) 211	(C) 5	(D) 90	(E) 210

PAPER FOLDING: 15 minutes

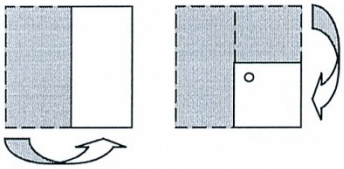
1.



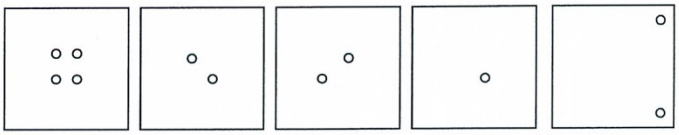
A. B. C. D. E.



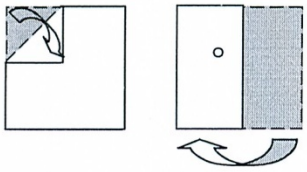
2.



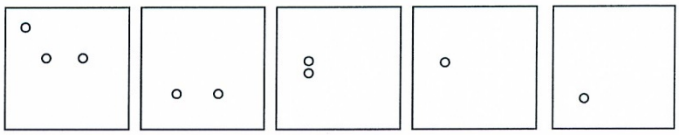
A. B. C. D. E.



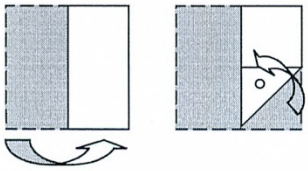
3.



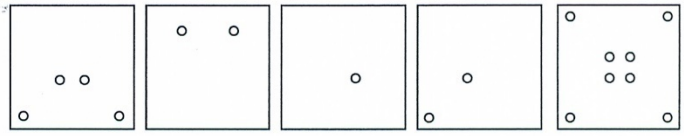
A. B. C. D. E.



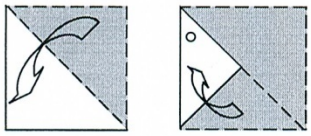
4.



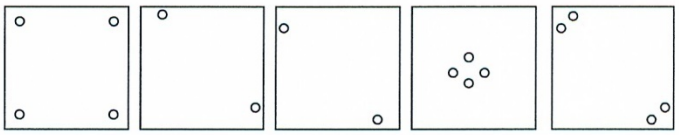
A. B. C. D. E.



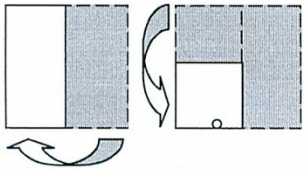
5.



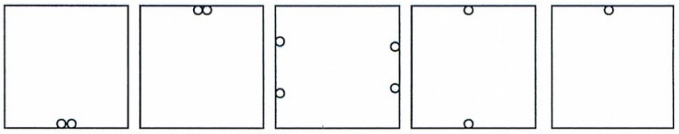
A. B. C. D. E.



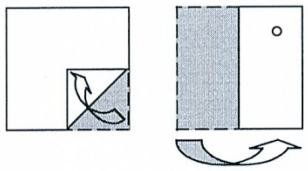
6.



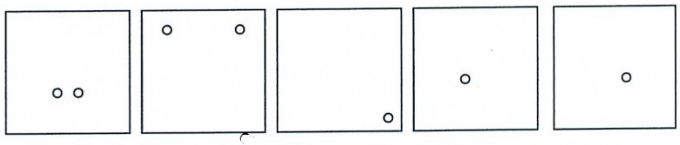
A. B. C. D. E.



7.



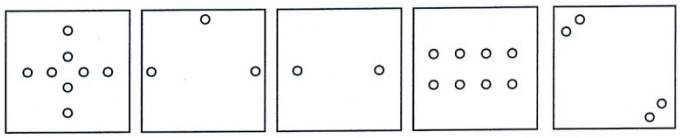
A. B. C. D. E.



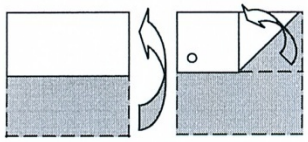
8.



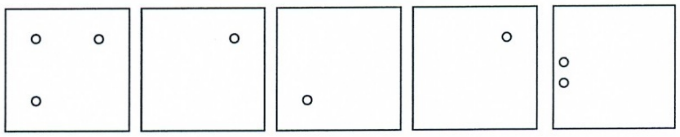
A. B. C. D. E.



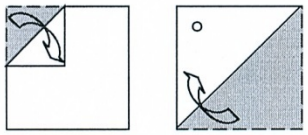
9.



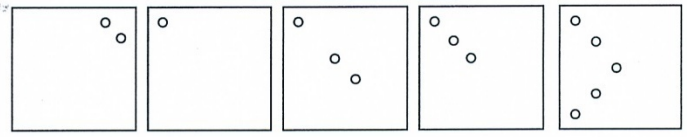
A. B. C. D. E.



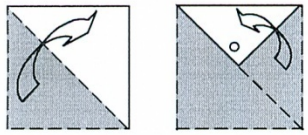
10.



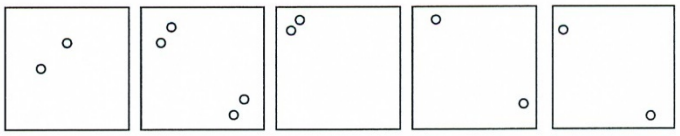
A. B. C. D. E.



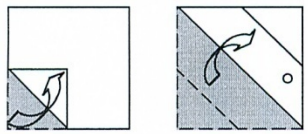
11.



A. B. C. D. E.



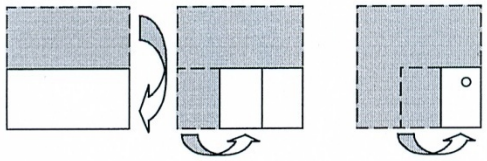
12.



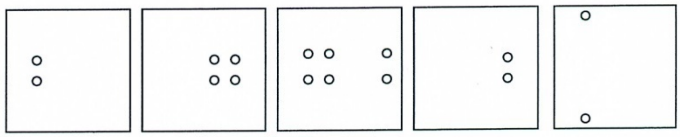
A. B. C. D. E.



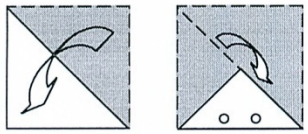
13.



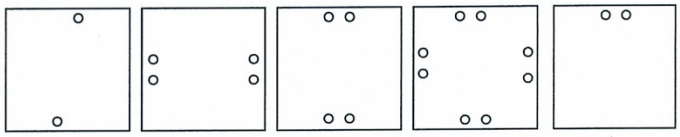
A. B. C. D. E.



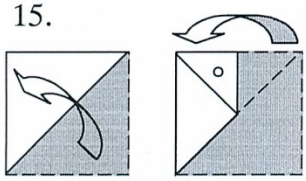
14.



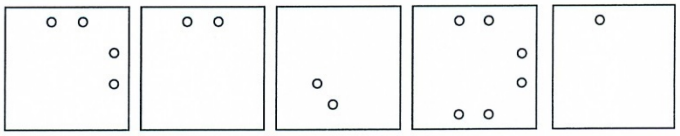
A. B. C. D. E.



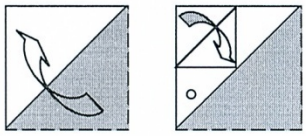
15.



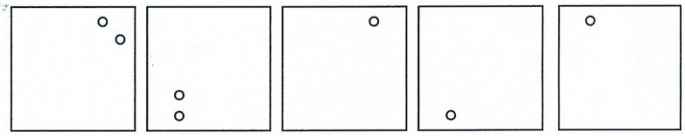
A. B. C. D. E.



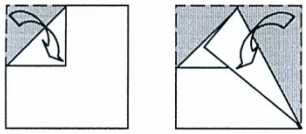
16.



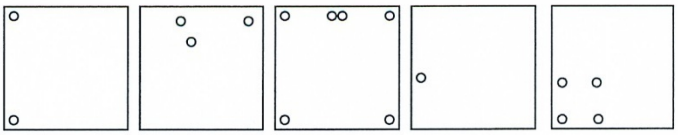
A. B. C. D. E.



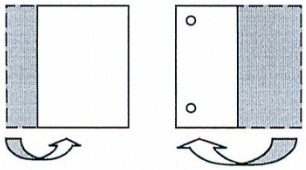
17.



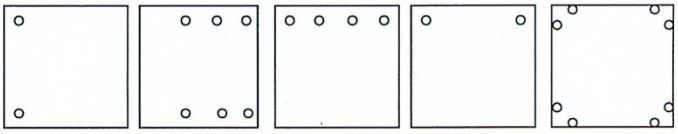
A. B. C. D. E.



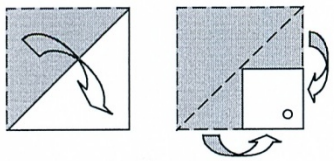
18.



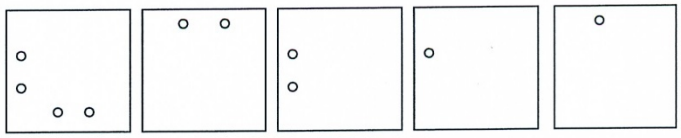
A. B. C. D. E.



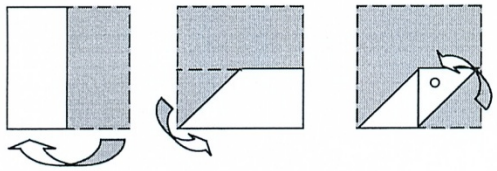
19.



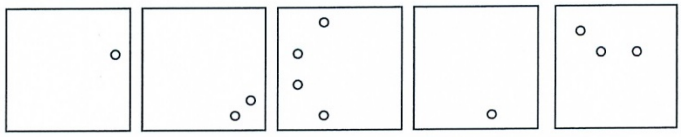
A. B. C. D. E.



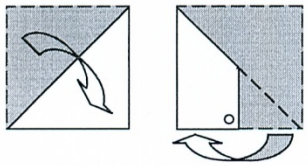
20.



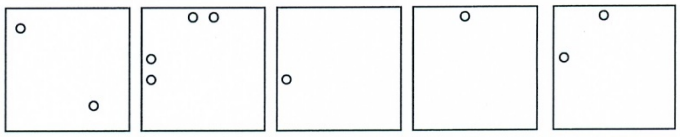
A. B. C. D. E.



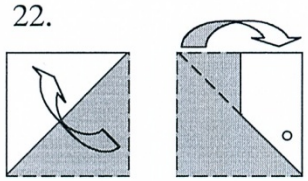
21.



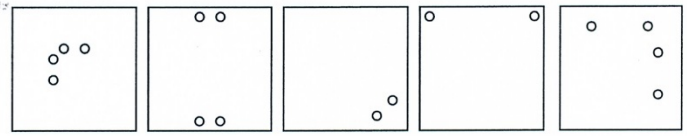
A. B. C. D. E.



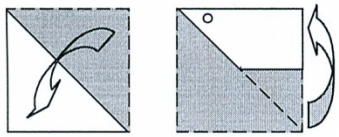
22.



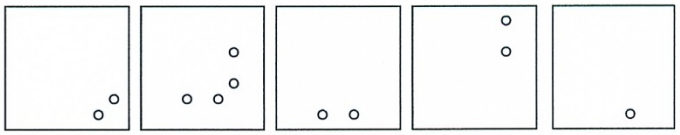
A. B. C. D. E.



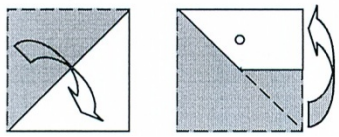
23.



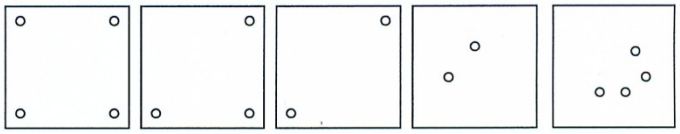
A. B. C. D. E.



24.



A. B. C. D. E.



READING: 40 minutes

The size of screwdrivers is measured by the length of the blade and not the overall length of the screwdriver. The diameter of the blade increases as the length of the blade increases. Also, the tip is wider and thicker as the length of the blade increases. Finally, the slots in the screw heads are wider as the screw size increases. As a result, the size of the screwdriver can match the size of the screw. A screwdriver with the largest tip should be used for the screw. If the screwdriver is too small for the size of the screw, it will be difficult to turn the screw or the screw will be damaged. The screwdriver blade should always be kept perpendicular to the screw head. Also, the screwdriver must be pushed as it is turned to hold it in the screw slot. Hammering on the screwdriver should be avoided. Hammering on the screwdriver usually means the wrong tool is being used, rather than a chisel, punch, or nail set.

1. The size of the screwdriver is measured by the
   A. width of the screwdriver
   B. width of the blade
   C. length of the blade
   D. weight of the blade

2. As the length of the blade increases, the diameter of the blade
   A. decreases
   B. increases
   C. remains the same
   D. depends on the material

3. As the length of the blade increases, the width of the tip is
   A. narrower and thicker
   B. wider and thinner
   C. remains constant
   D. wider and thicker

4. For a very large screw, the slot in the screw head would be
   A. large
   B. small
   C. straight
   D. narrow

5. The size of the screwdriver chosen for a particular screw should be
   A. small
   B. narrow
   C. large
   D. long

6. When fastening a screw, the screwdriver should be
   A. as small as possible
   B. at an angle to the crew head
   C. hit with a hammer
   D. pushed as it is turned

Several types of girders are used in steel fabrication. The most common types are the inverted girder, the beamlike girder, and the box girder. Girders are built-up beams and are fabricated in many different sizes from plate stock. The inverted girder is used mainly for supporting long span roofs, such as warehouses and factories. The beamlike girder is used for highway and railroad bridges, which are sometimes as long as 120 feet. The box girder is also used in highway and railroad applications, with the ability to hold heavy loads. Girders usually have connection stiffeners on both sides and bearing stiffeners at stress points. Shoe plates are used at the stress points where the girder will rest on the piers. From the above information, it is necessary to use the correct type of girder for specific purposes or buildings.

7. For steel fabrication, one of the most common types of girder is called
   A. straight
   B. in-line
   C. round
   D. box

8. Girders are fabricated from
   A. steel
   B. plate stock
   C. wire
   D. aluminum

9. An inverted girder would be used in the construction of a
   A. swimming pool
   B. factory building
   C. storage shed
   D. garage

10. For railroad bridges, the best type of girder is a
   A. box girder
   B. truncated girder
   C. inverted girder
   D. fabricated

11. At stress points, girders usually have stiffeners that are called
   A. tapered
   B. bearing
   C. vertical
   D. box

12. From the information above, the type of girder chosen depends on
    A. preference
    B. location
    C. the type of building
    D. tools for installation

If two bodies with different temperatures can exchange heat with one another, they will finally both acquire the same temperature at some intermediate value between their original temperatures. Two different types of heat transfer will be discussed here: convection and radiation. In the case of convection, heat transfer is effected by the flow of a gas or liquid caused by local heating and the effect of gravity. For instance, this could occur when a room is heated by warm air rising from a stove or a radiator. Thermal radiation, on the other hand, is independent of heat-transmitting matter and can even be achieved through a vacuum. Every hot body emits rays (electromagnetic waves) which are located in the infra-red portion of the spectrum. These waves have wavelengths longer than those of visible light rays. Bodies which absorb a large amount of radiation also re-emit a large amount of radiation. On the other hand, a suitably transparent body will, in general, allow radiant heat to pass and will undergo only a slight rise in temperature.

13. If two liquids of different temperatures are combined, the temperature of the resulting solution will be

    A. higher than both solutions individually.
    B. an average of both solutions.
    C. lower than both solutions individually.
    D. the reciprocal of the temperatures squared.

14. Light rays consist of electromagnetic waves which are _____ than infra-red waves.
    A. shorter
    B. longer
    C. wider
    D. narrower

15. When a transparent body is heated radiantly, there will be a _____ rise in temperature.
    A. large
    B. small
    C. intermediate
    D. zero

Pliers are holding tools which increase the usefulness of your hands by increasing gripping power or for holding hot or cold objects. Plier sizes are determined by the length of the pliers. The larger the size, the more powerful the pliers. Adjustable joint pliers can be set so that the jaws pivot, changing the maximum opening of the jaws. This allows the jaw opening to match the size of the object to be gripped. Linemen's side cutting pliers are useful for bending or cutting wire. Diagonal cutting pliers have slopping cutting edges but have no jaws for gripping. They are useful in mechanical work requiring cotter pins. Vise grip pliers are a multi-purpose tool that are both pliers and a clamp or vise. When the handles are closed together the jaws close on the work.

16. Plier sizes are determined by the
    A. weight of the pliers
    B. length of the pliers
    C. length of the jaws
    D. weight of the pliers

17. For work on large, heavy duty bolts, the pliers should be
    A. smaller
    B. wide
    C. larger
    D. size does not matter

18. Ajustable pliers can be used on _____bolts.
    A, small
    B. large
    C. any size
    D. steel

.

19. Lineman's side cutting pliers would be useful for a
    A. iron worker
    B. sheet metal worker
    C. pipefitter
    D. electrician
    E. all of the above

20. Vise grips are multi-purpose tools that work as
    A. pliers
    B. Clamps
    C. Vises
    D. both piers & clamps or vises
    E. none of the above

Devices known as valves, cocks, taps, and faucets are used for controlling the flow of liquids and gases. "Valve" is the general term. Therefore, the familiar water tap (or faucet) is technically a valve. A plug cock, on the other hand, is a simpler device in which the fluid passage is a hole in a rotatable plug which has a slightly tapered shape and fits in the body of the

cock. Opening and closing the cock is done by rotating the plug an angle of 90 degrees. Some degree of flow control is possible by setting the plug of the cock in an intermediate position. Much more accurate control of the rate of flow is provided by a valve, of which there are a number of different types. A feature common to most manually operated flow control valves is the stem. This is a screw spindle mounted by a hand wheel. The fluid (gas or liquid) flows through an opening whose edge forms the valve seat. The ordinary domestic water tap illustrates this principle. The end of the stem holds a disc which is usually provided with a replacement sealing washer to make the actual contact with the seat, thus stopping the flow of fluid. To open the tap, the disc is raised by rotating the hand wheel in a counterclockwise direction, so that the stem is screwed out of the valve body.

21. A waste tap is probably an example of a
    A. pipe.
    B. valve.
    C. plug.
    D. hand wheel.

22. Fluid passes through a plug cock by
    A. a valve.
    B. rotating the cock.
    C. rotating the plug 180 degrees.
    D. rotating the plug between 0 and 90 degrees.

23. A valve provides _____ control as compared to a cock.
    A. greater
    B. less
    C. the same
    D. an average amount of

24. The fluid in a valve flows through a(n) _____ which is usually sealed by a disc.
    A. opening
    B. faucet
    C. plug
    D. pipe

25. The ordinary domestic water tap is an example of a
    A. stem.
    B. faucet
    C. valve seat.
    D. plug.

26. With a water tap, water flow is stopped by a
    A. valve.
    B. valve spindle.
    C. screw spindle.
    D. disc with a washer.

27. To open a valve on a water tap, the
    A. disc is lowered.
    B. disc turns clockwise.
    C. handle is turned counterclockwise.
    D. handle is turned clockwise.

Slope or slant roofs are commonly used for homes, small commercial buildings and churches. The distance from one roof eave to the next roof eave is called the span. Half the distance of the span is the run. There are many designs for steep slope roofs. For example, a gable roof is shaped like an inverted V. It forms a ridge where the two sloped sections join. A hip roof slopes in four directions from a common ridge. A simple hip roof joins in a peak. A gambrel roof slopes in two directions from a common ridge with a break in the top sloping fields. Other types of steep sloped roofs are shed roofs and mansard roofs.

28. The slant of a roof can be described with the term
    A. rise
    B. slope
    C. degree
    D. hip

29. The distance from one roof eave to the next is called
    A. hip
    B. pitch
    C. span
    D. run

30. Half the span of a roof is called the
    A. rise
    B. run
    C. pitch
    D. slope

31. Examples of steep sloped roofs are
    A. gable roofs
    B. hip roofs
    C. gambrel roof
    D. shed roofs
    E. all the above

Practice Problems: Answer Key

Industrial Numerical Computation Key

1. **D**
2. **E**
3. **B**
4. **D**
5. **E**
6. **C**
7. **A**
8. **E**
9. **A**
10. **A**
11. **C**
12. **B**
13. **A**
14. **E**
15. **D**
16. **B**
17. **D**
18. **E**
19. **A**
20. **D**

21. E

22. A

23. B

24. D

25. E

26. D

27. C

28. B

29. A

30. E

Numerical Reasoning

1. **B**
2. **E**
3. **D**
4. **A**
5. **E**
6. **C**
7. **B**
8. **A**
9. **D**
10. **B**
11. **B**
12. **B**
13. **E**
14. **D**
15. **C**
16. **A**
17. **B**
18. **E**
19. **D**
20. **E**

Paper Folding

1. **B**
2. **A**
3. **A**
4. **A**
5. **E**
6. **C**
7. **B**
8. **A**
9. **E**
10. **B**
11. **B**
12. **D**
13. **C**
14. **D**
15. **A**
16. **A**
17. **B**
18. **A**
19. **A**
20. **C**
21. **B**

22. C

23. A

24. D

Industrial Reading Key

1. **C**
2. **B**
3. **D**
4. **A**
5. **C**
6. **D**
7. **D**
8. **B**
9. **B**
10. **A**
11. **B**
12. **C**
13. **B**
14. **A**
15. **B**
16. **B**
17. **C**
18. **C**
19. **D**
20. **D**
21. **B**
22. **D**

23. A

24. A

25. A

26. D

27. C

28. B

29. C

30. B

31. E